

Fig. 1: Comparison of conduction (dotted line) and total losses (solid line) of boost MOSFET (blue) and diode (red lines), respectively, as a function of  $R_{DS(on)}$  or slope of the ohmic forward characteristic, respectively. The ampere values indicate the rated current of diodes with a resistivity as given on the X-axis

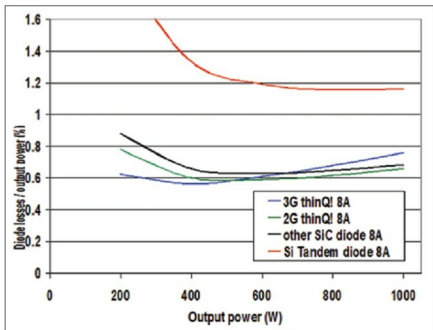


Fig. 2: Contribution of diode losses to the efficiency (diode losses/output power) as a function of output power in a 1000W PFC stage at 90V input and 130kHz switching frequency

wide input voltage range running at 130 kHz. Fig. 1 shows the conduction losses (in dotted lines) and total losses both for boost MOSFET and diode in this example.

An ideal match here would be 75 mOhm for the PFC MOSFET and the 8A third-generation SiC Schottky diode as both devices lead to an optimum, which is close to the minimum of losses but on the right hand side

of this optimum (which is important with respect to cost-effectiveness).

Now taking this 8A diode and varying the load from 20 per cent to 100 per cent gives a parabolic behaviour of the losses relative to the rated output power. Capacitive losses, which remain in their absolute value independent of the output power, increase inversely proportional to the output power as shown in the graph.

## Features

1. Lowest switching losses due to lowest  $Q_c$  ( $Q_{rr}$ ) for any current rating in the market
2. Fully surge-current stable, high reliability and ruggedness
3. Lower cost

Vice versa, conduction losses, which go absolutely with the square of the current or output power, change to a linearly increasing dependency in a graph where the relative losses vs output power are plotted.

Fig. 2 shows the result.

At light load, third-generation SiC diodes give 0.26 per cent better efficiency than competing 8A SiC Schottky barrier parts. Whereas, at full load, a small efficiency penalty of 0.08 per cent is seen. Due to better thermal coupling, there is practically no increase in junction temperature. Furthermore, thermal runaway is avoided by the merged p-n/Schottky concept. At light load, however, efficiency is worth a lot as it is becoming increasingly difficult to meet the ever-demanding efficiency targets there. At input voltages higher than 90V, the advantage of this optimisation is becoming even bigger.

Additionally, there are SiC Schottky diodes available in not only the TO-220 package (real two-pin version) but also the DPAK package for high-power-density surface-mount designs.

## Applications and benefits

SiC diodes can be used in SMPS (continuous-conduction-mode power-factor correction), LCD, PDP, lighting, motor drive, UPS and solar applications.

Benefits include system efficiency improvements at light and medium loads, lower system costs due to reduced cooling requirements, and broadest range of current ratings and lower costs/ampere for cost-effective performance improvements. Also, SiC diodes enable higher-frequency designs and increased power-density solutions. ●